Engineering Chemistry · Unit_05 · Engineering Chemistry - Lab Component

DRAFT — UE26CY151A · Unit_05 — Engineering Chemistry - Lab Component · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_05 · Engineering Chemistry - Lab Component

1 UE26CY151A · Unit_05: Engineering Chemistry - Lab Component

This unit master organizes source material for Engineering Chemistry, Unit_05, officially titled Engineering Chemistry - Lab Component. The official catalogue includes this unit, but the current course bucket has no matching unit folder or unit-specific lecture files. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26CY151A · Unit_05: Engineering Chemistry - Lab Component

Official catalogue entry: UE26CY151A, Unit_05 (Engineering Chemistry - Lab Component). Bucket folder present: no. Videos: 0; slide decks: 0. No unit-specific current lecture sources are present in the bucket.

EV0000001

2 Historical question-source pages · unit mapping candidates

Study focus: Historical question-source pages · unit mapping candidates. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000002 · EV0000003

SHORT-NOTE RECAP

Study notes · Historical question-source pages · unit mapping candidates

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Archive page 1 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Department of Science and Humanities ENGINEERING CHEMISTRY

Archive page 2 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY • INTRODUCTION • COMPUTATIONAL METHODS • APPLICATIONS

Archive page 3 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY Introduction Computational chemistry is a set of techniques for the investigating chemical problems on a computer rather than using chemicals. It uses the results of theoretical chemistry incorporated into efficient computer programs to calculate the properties of a molecule. The first theoretical investigation was done by Walter Heitler and Fritz London in 1927 lecture_notes.dvi

Archive page 4 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY 1998 Walter Kohn : For the development of density functional theory. John A. Pople : For the development of computational methods in quantum chemistry.

Archive page 5 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY 2013 For the development of computer based methods to model complex systems.

Archive page 6 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY Objective of computational chemistry: To solve chemistry problems by simulating chemical systems (molecular, biological, materials) to provide reliable, accurate and comprehensive information at an atomic level. Advantages of computational chemistry: ☒Calculations are easy to perform than carrying experiments. ☒Calculations are less costly, whereas experiments are becoming more expensive ☒Calculations are safe than performing experiments. lecture_notes.dvi

Archive page 9 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY Drug discovery: Computational chemistry plays a pivotal role in design and screening of potential drug candidates by simulating the interactions between the drug molecules and biological targets. Paracetamol All these spectra are generated using Gaussion 16

Archive page 11 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Department of Science and Humanities ENGINEERING CHEMISTRY

Archive page 13 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY • Molecular mechanics: ☒Molecular mechanics programs use equations based on classical physics to calculate force fields. ☒It is based on the model of a molecule as a collection of balls(atoms) held together by springs(bonds). ☒Its too fast like a very large molecules like steroids can be optimized in seconds and allows geometry optimization .

Archive page 14 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY • Molecular dynamics: ☒Molecular dynamics is a molecular mechanics program designed to mimic the movement of atoms within a molecule. The laws of motion to molecule is the base for it. ☒Molecular dynamics can be carried out on a molecule to generate different conformation which on energy minimization. Staggered vs Eclipsed Conformations of Ethane - Master Organic Chem

Archive page 15 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY • Semi empirical methods: ☒Semi empirical calculations are based on Schrödinger equation. ☒It represents a middle road between the mostly qualitative results available from molecular mechanics and the high computationally demanding quantitative results from Ab initio methods. ☒Capable of calculating transition states and excited states.

Archive page 16 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY Ab initio calculations: ☒Ab initio calculations are based on Schrödinger equation and it describes, among other things, how the electrons in a molecule behave. ☒The wave function is a mathematical function that can be used to calculate the electrons distribution.

Archive page 17 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY • Density functional theory ☒DFT are like Ab initio and semi empirical calculations, based on Schrödinger equation. ☒However, unlike the other two methods, DFT does not calculate a conventional wave function , but rather derives the electron distribution directly. ☒Density functional calculations are usually faster than Ab initio, but slower than semi empirical method.

Archive page 19 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY ☒Computational chemistry plays a pivotal role in advancing our understanding of chemical systems and accelerating scientific discovery. ☒Its advantages in cost-effectiveness, efficiency, safety, accessibility, versatility, and accuracy make it an indispensable tool in modern research and development. ☒Computational chemistry is not a replacement for experimental studies, but plays an important role in enabling chemists to design and develop the desired chemicals/ products. Conclusion. Ref. Book: Introduction to computational chemistry by Frank Jensen, Wiley publications

Archive page 23 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Hydrogen Energy • The Hydrogen energy has been identified as one of potential sources of fuel in the state-of-art technological era • Hydrogen production, storage and transportation has become the focus of science and technology • Sustainable production of Hydrogen from renewable feedstocks and electrolysis of water has taken lead role in technology development • More emphasis is provided to understanding and engineering the production and storage of Hydrogen produced by Electrolysis method 3 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage

Archive page 25 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Hydrogen Economy • Hydrogen used as fuel reduces pollution • It is carbon free energy source • Hydrogen economy is a vision of an energy delivery infrastructure based on hydrogen as a carbon-free energy carrier • It provides • alternative energy economy in the form of a parallel power and transportation infrastructure • technical solutions for energy-efficient hydrogen production, storage technology and delivery infrastructure 5 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage

Archive page 28 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. • Photolysis • Processes use light energy to split water into hydrogen and oxygen • One of the potential for sustainable hydrogen production • Low environmental impact • Yet in early stage of research • Electrolysis • uses electricity to split water into hydrogen and oxygen • Electrolysis reaction takes place in a unit called an electrolyzer • Method is promising option for carbon-free hydrogen production from renewable and nuclear resources. 8 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage

Archive page 30 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. • Shift reforming (Water Gas shift reactions) CO + H2O → CO2 + H2 • Catalyst: Fe3O4/Cr2O3 • Exothermic reaction (ΔHo = -41.0 KJ/mole) • Favoured at low temperature (350oC) • H2 produced will have traces of Carbon monoxide as impurities • Purification • CO2 is removed to produce pure Hydrogen • Traces of CO is removed by methanation (350 – 450oC) • CO + 3H2O ⊠CH4 + H2O 10 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage

Archive page 31 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Green Hydrogen refers to production of Hydrogen gas from renewable resources Green Hydrogen production will help us reach both the momentum and promise for greener mobility and industry Hydrogen production through Electrolysis produces clean energy with pollution free process by using electricity Hydrogen production by electrolysis can be achieved by three methods. 1. Alkaline Electrolysis 2. Proton Membrane Electrolysis 3. High temperature Electrolysis Green Hydrogen production 11 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage

Archive page 32 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. • In Alkaline electrolyte process • aqueous solution containing either NaOH or KOH used • typical concentration of 20-40 wt % • operation temperatures are between 343 – 363 K • operating pressure up to 3 MPa • Charge carrier: OH- • Anodic materials: Ni and its alloys • Cathodic materials: Cd, Pb, Cu, Ag, Pt, Pd. • Electrolyte materials : KOH, NaOH • Catalyst materials: Ni-Zn, Ni-Co-Au, RuO2, LaCoO3 12 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage

Archive page 34 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. • Zero emission of CO2 (if this process combined with renewable energy sources like electricity form solar or wind energy) • Alkaline water electrolysis is conventional methods with easiest, simplest, and suitable methods for Hydrogen production • Electrolysis requires needs Direct current supply • Electrolyser split water into Hydrogen and Oxygen gas 14 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage

Archive page 35 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. • Splitting of water is very small approximately 7- 10 moles/ litre • because pure water is the very poor conductor of electricity hence acid (H2SO4 in acid electrolyzer) or base (KOH or NaOH in Alkaline electrolyzer) is used to improve the conductivity • Solutions split into ions which improving conductivity of the electrolyte. • The Reaction of water splitting is as follows: • At Anode: 2OH- ☒½ O2 + H2O + 2 e- • At Cathode: 2H2O + 2e- ☒H2 + 2OH- • The basic equation of water splitting is represented as: 2H2O (liquid) + Energy ☒2H2(gas) + O2 (gas) 15 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage

Archive page 36 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Conclusion: • The Hydrogen energy has been identified as one of potential sources of fuel, Production of Green Hydrogen is the key focus to meet the sustainable development. • Hydrogen is not available as a pure gas, it must be produced from other compounds like fossil fuels, biomass, electrolysis, etc., • Electrolysis of water to produce hydrogen is one of the most viable technology • Hydrogen evolution takes place at cathode materials, whereas oxygen evolution takes place at anode materials. • Catalysts improve the performance of the electrode materials to generate high and pure hydrogen gas. 16 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage

Archive page 39 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Module content: • Hydrogen Storage • Methods of Storage 2 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage

Archive page 40 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Hydrogen Energy • The Hydrogen storage is crucial for effective utilization of Hydrogen for its versatile applications. • Methods used for hydrogen storage: • Liquid form • Gaseous • Solid State 3 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage

Archive page 41 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. • Hydrogen Storage in Metal-Organic Frameworks: A Review - research journal (gyanvihar.org) 4 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage

Archive page 42 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. 5 • Hydrogen can be physically stored as either a gas or a liquid • Storage of hydrogen in the form of gas usually requires high-pressure tanks (350–700 bar tank pressure) • Storage of hydrogen as a liquid requires cryogenic temperatures because the boiling point of hydrogen is −252.8°C at an atmospheric pressure • Hydrogen can also be stored on the surfaces of solids (adsorption) or within solids (absorption) 5 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage very low

Archive page 43 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Solid state hydrogen storage: • Depending on the type of hydrogen being stored in host materials, solid-state hydrogen storage can be classified into two categories: • Chemisorption of hydrogen atoms • Physisorption of intact hydrogen molecules • In

chemisorption (absorption into matter), binding of hydrogen atoms occur into the surface • In case of physisorption (adsorption on surface).The Hydrogen molecule is bound onto the surface of adsorbent 6 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage

Archive page 1 from 5__1pIeTzocnU.pdf; unit match candidate score 1.0; source text/OCR may be partial. 11/11/24, 11:34 AM ESA -JAN 2024 - UE23CY151A (set- 2) about:blank 1/7 (6.0 Marks) (6.0 Marks) (6.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - JAN 2023 UE23CY151A - Engineering Chemistry Total Marks : 100.0 1.a. For a rotating diatomic molecule, i) Derive an expression for moment of inertia 'I' for the molecule. ii) If B =10 cm , give the rotational energy levels of this molecule, in cm for rotational quantum numbers J=0,1,2,3, and draw an energy level diagram for the same molecule up to J = 3. 1.b. 1.c. (i) Draw a neat labeled phase diagram of the Carbon dioxide system specifying the different areas and lines. Give temperature and pressure values at the triple point and critical point. (ii) What is the Reduced phase rule? Why is general Gibb's Phase rule not applied to the 2-component system? -1 -1

EV0000002 · EV0000003 · EV0000004 · +26 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Extracted historical pages: Chem_Extra_Topics__1-HfOeZNwo.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Extra/Chem_Extra_Topics__1-HfOeZNwo.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 5__1pIeTzocnU.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/5__1pIeTzocnU.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. Unit_05 is listed in the official catalogue but has no matching source folder in PESU_Materials.

2. No current video or slide materials were available for this catalogue-only unit; archive excerpts appear only when the unit-match candidate is strong.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.
SRC00002	28 covered · 0 excluded · 0 gaps	q1: Original archive PDF page 1; candidate unit association is text-based and unconfirmed. q2: Original archive PDF page 2; candidate unit association is text-based and unconfirmed. q3: Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
SRC00003	1 covered · 0 excluded · 0 gaps	q1: Original archive PDF page 1; candidate unit association is text-based and unconfirmed.

C Revision and retrieval practice